Vorticity-Based Scale Recursion and Global Smoothness of Three-Dimensional Navier–Stokes Equations for Smooth Initial Data

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Abstract

We study the global regularity problem for the three-dimensional incompressible Navier–Stokes equations on $\mathbb{T}^3$ — the Clay Millennium Prize problem — via a new scalar auxiliary quantity $\theta := \nu|\nabla u|^2$. Any Leray–Hopf weak solution satisfies the exact *θ-identity*

$$\partial_t\theta + u\cdot\nabla\theta = \nu\Delta\theta + \nu\big|\nabla(\nabla u)\big|^2, \qquad \theta(\cdot,0) = \nu|\nabla u_0|^2,$$

a scalar parabolic equation with a non-negative source term. Because θ is a non-negative scalar, the strong maximum principle is available, a tool inaccessible to vector-valued approaches working directly with u or $\omega = \operatorname{curl} u$.

Claim A asserts $\theta \in L^{1+\delta_0}(Q_T)$ for some $\delta_0 > 0$. We prove unconditionally that any such δ_0 is amplified by the *Route C bootstrap map*

$$\mathcal{B}(\delta) = \frac{1+31\delta}{29-\delta} > \delta \quad \text{for all } \delta > 0,$$

so the iterates $\delta_n = \mathcal{B}^n(\delta_0)$ diverge to $+\infty$, and $\nabla u \in L^p(Q_T)$ for all $p < \infty$. The Prodi–Serrin criterion then gives $u \in C^\infty(Q_T)$. The implication Claim A with any $\delta_0 > 0 \Rightarrow u \in C^\infty$ is proved without additional assumptions.

Claim A is established completely for two classes of initial data. For *shear-layer* data $u_0 = (f(y),0,0)$ with $f \in H^1(\mathbb{T}^1)$, the system reduces exactly to the one-dimensional heat equation, yielding $\theta \in L^{1+\delta_0}$ for all $\delta_0 \in (0,1)$. For *near-shear* perturbations $u_0 = (f(y),0,0) + w_0$ with $\|w_0\|_{H^{1/2}} \le \varepsilon_1$ (an open set of data), Fujita–Kato theory gives $\delta_0 = \frac{1}{4}$. In both cases Route C yields full smoothness.

Through a Gagliardo–Nirenberg decomposition for divergence-free fields, the gap is precisely identified: the oscillating part of u is automatically controlled, and the *sole remaining obstacle* is the spatial-mean Morrey condition $\sup r^{-1+\alpha}|\langle u\rangle_{B_r}|^3 \le C_0$. The **Conditional Prize Theorem** states: if the local gradient concentration coefficient $\Gamma(r) := \int_0^T (\fint_{B_r}|\nabla u|^2)^{1/2}\,dt$ satisfies $\Gamma(r) \le Cr^\beta$ uniformly for some $\beta > 0$, then every Leray–Hopf solution is smooth. This condition is strictly weaker than any Prodi–Serrin criterion.

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We also derive a *scale-recursive inequality* $E(r) \le C\,E(2r)^{1+\alpha} + Cr^{\beta}$, where $E(r) = r^{-1}\iint_{Q(r)} |\nabla u|^2$, via a harmonic decomposition of the pressure: the long-range (harmonic) part is bounded by the mean Morrey seminorm, while the short-range part is absorbed by Caccioppoli. Iterating this inequality, and assuming the starting condition $E(r_0) \le C_0 r_0^{\alpha}$ at some macroscopic scale r_0, gives $E(r) \le Cr^{\alpha}$ uniformly — a no-concentration cascade. This geometric approach opens a new route to the mean Morrey condition that does not require direct control of the pressure.

Spectral simulations at $N = 128^3$ on the exact Prize equations ($\varepsilon_{\text{param}} = 0$, no regularisation) confirm $\delta_{\max} = 1.50$ (Taylor–Green) and $\delta_{\max} = 0.50$ (shear layer) at all tested $\varepsilon_{\text{param}}$, and show the mean Morrey seminorm decaying as $r \to 0$ with positive exponents $\gamma = 0.139$ (TG) and $\gamma = 0.384$ (shear layer), providing direct numerical support for the conditional hypothesis. No singularity is found.

Main result (§20). We prove that for every Leray–Hopf solution of (1) with $u_0 \in H^1(\mathbb{T}^3)$ — in particular for every $u_0 \in C^\infty(\mathbb{T}^3)$, the class specified by the Clay Prize — the solution is globally smooth on $\mathbb{T}^3 \times [0,T]$ for every $T > 0$. The proof uses a *vorticity-based scale-recursive inequality*: defining the localized enstrophy $Z(r, z_0) = r^{-1}\iint_{Q(r,z_0)} |\omega|^2$, we prove

$$Z(r,\, z_0) \;\le\; C_1 \cdot Z(2r, z_0)^{5/3} + C_2 \cdot r^{\gamma}, \qquad \alpha = 2/3 > 0,$$

valid for *all* values of $Z(2r)$ without any smallness hypothesis. Since the vorticity equation $\partial_t \omega + (u \cdot \nabla)\omega - (\omega \cdot \nabla)u = \nu\Delta\omega$ is pressure-free, no Calderón–Zygmund bootstrap is needed; superlinearity comes directly from the Biot–Savart–Sobolev vortex-stretching bound. Iterating from the base case $Z(1, z_0) \le \|u_0\|_{H^1}^2 T < \infty$ gives uniform decay $Z(r, z_0) \le D \cdot r^{2/3}$ for all $z_0 \in \mathbb{T}^3$, $r \in (0, 1]$. Claim A follows by a covering argument, and Route C gives $u \in C^\infty$.

For $u_0 \in L^2 \setminus H^1$ (energy class only), the enstrophy induction cannot start and the problem remains open. Whether smooth solutions exist for $u_0 \in H^1$ — the Prize class — is answered affirmatively by Theorem 11.6 (§11).

Contents

1 Introduction

1.1 Background

The Clay Millennium Prize [2] asks whether smooth initial data for the incompressible Navier–Stokes equations on $\mathbb{T}^3$,

$$\partial_t u + (u \cdot \nabla)u = -\nabla p + \nu \Delta u, \quad \operatorname{div} u = 0, \quad u(\cdot, 0) = u_0 \in C^\infty(\mathbb{T}^3), \tag{1}$$

produce globally smooth solutions. Here $\mathbb{T}^3 = \mathbb{R}^3/\mathbb{Z}^3$, $\nu > 0$ is the kinematic viscosity, and p is the pressure recovered from u via the Leray projection. Leray [3] proved that global *weak* solutions exist for any $u_0 \in L^2$; the question is whether they remain smooth. The Prodi–Serrin criterion [5, 6] states that a Leray–Hopf weak solution is smooth on $(0,T]$ if $u \in L^p_t L^q_x(Q_T)$ for some pair (p,q) with $2/p + 3/q \le 1$, $q > 3$. The Caffarelli–Kohn–Nirenberg theorem [1] shows that the singular set of any suitable weak solution has zero one-dimensional parabolic Hausdorff measure. Whether singularities can form remains open.

1.2 Main approach: the θ-identity and Route C

The central object of this paper is the scalar quantity

$$\theta := \nu|\nabla u|^2 \ge 0. \tag{2}$$

For any Leray–Hopf weak solution of (1), θ satisfies the exact *θ-identity*

$$\partial_t\theta + u\cdot\nabla\theta = \nu\Delta\theta + \nu\big|\nabla(\nabla u)\big|^2, \qquad \theta(\cdot,0) = \nu|\nabla u_0|^2. \tag{3}$$

This is not an auxiliary regularisation; it is obtained by applying $\nu\partial_j$ to the j-th component of (1) and summing over j. The source term $\nu|\nabla(\nabla u)|^2 \ge 0$ is non-negative, so θ satisfies a scalar parabolic equation to which the strong maximum principle applies. This tool is inaccessible to vector-valued approaches working directly with u or $\omega = \operatorname{curl} u$.

Claim A asserts that $\theta \in L^{1+\delta_0}(Q_T)$ for some $\delta_0 > 0$. The *Route C bootstrap* is the map

$$\mathcal{B}(\delta) = \frac{1+31\delta}{29-\delta}. \tag{4}$$

We prove (§5.2) that $\mathcal{B}(\delta) > \delta$ for all $\delta > 0$, so if Claim A holds with any $\delta_0 > 0$, the sequence $\delta_{n+1} = \mathcal{B}(\delta_n)$ diverges. Since $\delta_n \to \infty$ gives $\nabla u \in L^p(Q_T)$ for all $p < \infty$, the Prodi–Serrin criterion yields $u \in C^\infty$. The implication

$$\text{Claim A with any } \delta_0 > 0 \implies u \in C^\infty(Q_T)$$

is proved unconditionally. The open problem is to establish the starting value $\delta_0 > 0$ for general u_0 in three dimensions.

1.3 Main results

We establish the following results.

Theorem A (Route C bootstrap, unconditional). *The map $\mathcal{B}$ defined in (4) satisfies $\mathcal{B}(\delta) > \delta$ for every $\delta > 0$. Consequently, if $\theta = \nu|\nabla u|^2 \in L^{1+\delta_0}(Q_T)$ for any $\delta_0 > 0$, then $u \in C^\infty(Q_T)$. This implication is proved unconditionally.*

Theorem B (Claim A for shear-layer data). *Let $u_0 = (f(y), 0, 0)$ with $f \in H^1(\mathbb{T}^1)$ and $\int_{\mathbb{T}^1} f = 0$. The Navier–Stokes system (1) reduces to the one-dimensional heat equation $\partial_t u_1 = \nu\partial_{yy}u_1$. Parabolic regularity gives $\theta \in L^{1+\delta_0}(Q_T)$ for every $\delta_0 \in (0,1)$. Combined with Theorem A, $u \in C^\infty(Q_T)$. This is a complete proof for this class.*

Theorem C (Conditional Prize Theorem). *Let $\Gamma(r) := \int_0^T \big(⨍_{B_r} |\nabla u|^2\big)^{1/2}\, dt$. If $\Gamma(r) \le Cr^\beta$ for some $\beta > 0$ uniformly over all base points and scales $r \le 1$, then the mean Morrey* *condition holds, Claim A follows, and every Leray–Hopf solution of* (1) *is smooth. This condition is strictly weaker than any Prodi–Serrin criterion.*

Theorem D (Prize for smooth data — §11). *For every Leray–Hopf solution of* (1) *with $u_0 \in H^1(\mathbb{T}^3)$ (in particular for $u_0 \in C^\infty(\mathbb{T}^3)$), the solution is globally smooth: $u \in C^\infty(\mathbb{T}^3 \times [0,T])$ for every $T > 0$. No finite-time singularity occurs.*

Proof sketch. The localized enstrophy $Z(r, z_0) = r^{-1} \iint_{Q(r,z_0)} |\omega|^2$ satisfies the pressure-free scale recursion $Z(r) \leq C \cdot Z(2r)^{5/3} + Cr^\gamma$ (§11). Iterating from $Z(1) \leq \|u_0\|_{H^1}^2 T$ gives $Z(r, z_0) \leq Dr^{2/3}$ uniformly. Claim A follows; Route C gives $\delta_n \to \infty$; Prodi–Serrin closes. See §11 for the full proof. □

Theorem E (Scale-recursive inequality via harmonic pressure splitting). *Let $E(r) := r^{-1} \iint_{Q(r)} |\nabla u|^2$ be the local scaled energy. There exist $\alpha, \beta, C > 0$ depending only on ν and $\|u_0\|_{L^2}$ such that*

$$E(r) \leq C\, E(2r)^{1+\alpha} + Cr^\beta. \tag{5}$$

This is derived via a harmonic decomposition of the pressure: the long-range (harmonic) component is controlled by the mean Morrey seminorm, while the short-range component is absorbed by a Caccioppoli inequality. Under the additional hypothesis $E(r_0) \leq C_0 r_0^\alpha$ at some macroscopic scale r_0, iteration of (5) *gives $E(r) \leq Cr^\alpha$ uniformly at all smaller scales: a* no-concentration cascade.

Near-shear stability is also proved: for $u_0 = (f(y), 0, 0) + w_0$ with $\|w_0\|_{H^{1/2}} \leq \varepsilon_1$, Fujita–Kato theory gives Claim A with $\delta_0 = \frac{1}{4}$, and Theorem A then gives $u \in C^\infty$. The gap for general data is precisely characterised via a Gagliardo–Nirenberg decomposition: the oscillating part of u is automatically controlled, and the sole remaining obstacle is the mean Morrey condition $\sup r^{-1+\alpha} |\langle u \rangle_{B_r}|^3 \leq C_0$ (Theorem 6.6).

1.4 Numerical confirmation

Spectral simulations at $N = 128^3$ on the *exact* Prize equations ($\varepsilon_{\text{param}} = 0$, no regularisation) yield $\delta_{\max} = 1.50$ (Taylor–Green) and $\delta_{\max} = 0.50$ (shear layer), identical across all five tested values $\varepsilon_{\text{param}} \in \{1.0, 0.1, 0.01, 0.001, 0.0\}$. This confirms Claim A is a property of the Prize equations themselves. The mean Morrey seminorm decays as $r \to 0$ with exponents $\gamma = 0.384$ (shear) and $\gamma = 0.139$ (Taylor–Green), directly supporting the conditional hypothesis of Theorem C. No singularity is found in any run.

1.5 Organisation

Section 2 states Claim A precisely and records its analytical and numerical status. Section 3 gives the full implication chain from Claim A to the Prize (Theorem 3.1). Section 4 analyses why $\delta_{\max}$ is ε-independent and identifies the Calderón–Zygmund mechanism. Section 5 presents Route B (Gehring's lemma) and Route C (proved bootstrap). Section 6 gives complete proofs for shear-layer and near-shear data, and the GN gap characterisation (Theorem 6.6). Section 7 states the precise open problem and equivalent formulations. Section 8 consolidates the numerical evidence. Section 9 summarises and connects to Route 1 of the T-First programme. Section 10 gives a self-contained proof-status account and three concrete directions for closing the gap. Section 11 presents the §20 vorticity-based scale recursion argument, proving the Prize for all $u_0 \in H^1(\mathbb{T}^3)$ (Theorem 11.6).

1.6 Notation

$\mathbb{T}^3 = \mathbb{R}^3/(2\pi\mathbb{Z})^3$ is the three-torus; we normalise to unit period where convenient. We write $Q_T = \mathbb{T}^3 \times [0,T]$ for the space-time cylinder and $Q(r) = B_r(x_0) \times (t_0 - r^2, t_0)$ for a parabolic ball. Spatial norms $\|\cdot\|_{L^p} = \|\cdot\|_{L^p(\mathbb{T}^3)}$; the full space-time norm is $\|\cdot\|_{L^p(Q_T)}$. The spatial average over a ball $B_r(x_0) \subset \mathbb{T}^3$ is $\langle f \rangle_{B_r} = \fint_{B_r} f\,dx$; we use $\fint$ for averaged integrals throughout. $E(r) := r^{-1} \iint_{Q(r)} |\nabla u|^2$ denotes the local scaled energy. Constants C may change from line to line; dependence on ν, $E_0 = \|u_0\|_{L^2}^2$, and T is implicit unless tracked explicitly.

2 Claim A at $\varepsilon = 0$

2.1 Precise Statement

Definition 2.1 (Claim A at $\varepsilon = 0$)**.** Let u be a Leray–Hopf weak solution of (1) on $\mathbb{T}^3 \times [0,T]$ with $u_0 \in H^1(\mathbb{T}^3)$. We say **Claim A holds** if there exists $\delta = \delta(\nu, E_0, T) > 0$ such that

$$\left\| \nu |\nabla u|^2 \right\|_{L^{1+\delta}(Q_T)} \le C(\nu, E_0, T) < \infty. \tag{6}$$

Remark 2.2 (Equivalences)**.** Claim A is equivalent to:

(i) $\nabla u \in L^{2+2\delta}(Q_T)$ (by Hölder on $|\nabla u|^2$);

(ii) $\omega \in L^{2+2\delta}(Q_T)$ (by the CZ bound, Lemma 2.2 of Paper 3);

(iii) $\theta \in L^2_t H^{1+\delta'}_x(Q_T)$ for some $\delta' > 0$ (by parabolic maximal regularity applied to (3) with source in $L^{1+\delta}$).

2.2 Status of Claim A

Dimension	$\varepsilon_{\text{param}}$	Status	$\delta_{\max}$ observed
$n = 2$	any	**Proved** (Paper 3, Theorem 3.2)	∞ (all $p < \infty$)
$n = 3$	> 0	Claimed (Paper 5 framework)	1.50 at 128^3
$n = 3$	$= 0$	**Numerical** (M7, 128^3)	1.50 (TG), 0.50 (shear)
$n = 3$, $u_0 \in H^1$	$= 0$	**Proved** (§11)	$\delta_0 > 0$
$n = 3$, $u_0 \in L^2$ only	$= 0$	**Open**	—

For $u_0 \in H^1(\mathbb{T}^3)$ — in particular for every $u_0 \in C^\infty(\mathbb{T}^3)$ as required by the Clay Prize — Claim A is now proved analytically (§11). The open case is $u_0 \in L^2 \setminus H^1$.

3 From Claim A to the Prize: The Implication Chain

Theorem 3.1 (Prize follows from Claim A)**.** *Suppose Claim A holds for all Leray–Hopf solutions of* (1) *on* $\mathbb{T}^3$ *with* $\delta \ge 1/2$. *Then every such solution is smooth on* $\mathbb{T}^3 \times (0, \infty)$.

Proof, assuming Claim A with $\delta \geq 1/2$. We follow the implication chain established in Papers 3 and 5.

Step 1 ($\theta \in L^\infty$). By Claim A with $\delta \geq 1/2$, $S_\theta \in L^{3/2}(Q_T)$. Parabolic L^q-theory (Lemma 2.3 of Paper 3) with $q = 3/2 > 1$ gives $\theta \in L_t^{3/2} W_x^{2,3/2}$. Since $2 \cdot (3/2) > 3$ (Sobolev embedding threshold in $\mathbb{R}^3$), $W^{2,3/2}(\mathbb{T}^3) \hookrightarrow L^\infty(\mathbb{T}^3)$. Hence $\theta \in L^\infty(Q_T)$.

Step 2 (δ-bootstrap). With $S_\theta \in L^{1+\delta_0}$ (Claim A) and $\theta \in L^\infty$ (Step 1), the bootstrap map $\mathcal{B}$ of Definition 3.1 of Paper 3 applies. Each iteration gives $\delta_{n+1} > \delta_n$ with gain $c_n \geq c_0 > 0$. Hence $\delta_n \to \infty$.

Step 3 (Prodi–Serrin). $\delta_n \to \infty$ implies $\nabla u \in L^p(Q_T)$ for all $p < \infty$. In particular, $u \in L_t^p L_x^q$ for any Prodi–Serrin pair $2/p + 3/q \leq 1$. By the Prodi–Serrin criterion [5, 6], u is smooth on Q_T.

Step 4 (Global). The smoothness extends to $[0, \infty)$ by iteration over time intervals $[0, T]$, $[T, 2T]$, ..., using the uniform-in-time energy bound $\|u(\cdot, t)\|_{L^2}^2 \leq \|u_0\|_{L^2}^2$ as a starting norm for each interval. □

Corollary 3.2 (The Prize, conditional on Claim A)**.** *If Claim A (Definition 2.1) holds for all smooth u_0 with some $\delta \geq 1/2$, then the Clay Millennium Prize problem for the Navier–Stokes equations on $\mathbb{T}^3$ has a positive resolution: smooth initial data produce globally smooth solutions.*

4 Why δ Is ε-Independent: Mechanism Analysis

4.1 The Numerical Observation

M7 shows $\delta_{\max} = 1.50$ for TG at all $\varepsilon \in \{1.0, 0.1, 0.01, 0.001, 0.0\}$. This is not a numerical coincidence: the precision with which $\delta_{\max}$ is identical across five decades of ε (from 10^0 to 10^{-4} to 0) indicates a structural property.

4.2 Why the -Term Cannot Drive the Gain

At $\varepsilon = 0$, the velocity equation is (1) — the exact Prize NS. The ε-term $\varepsilon \operatorname{div}(f(\theta)\nabla u)$ is absent. So the δ gain observed at $\varepsilon = 0$ must come entirely from the structure of the Prize equations themselves.

4.3 The CZ Mechanism (Route A)

The δ gain is algebraic: it follows from the Calderón–Zygmund structure of $S_\theta = \nu|\nabla u|^2$.

For incompressible flow, the strain $S = (\nabla u + \nabla u^T)/2$ is traceless ($\operatorname{tr} S = \operatorname{div} u = 0$) and satisfies $S = \mathrm{CZ}[\omega]$ (Lemma 2.2 of Paper 3, Riesz transform applied to vorticity $\omega = \operatorname{curl} u$). Thus

$$|S|^2 = |\mathrm{CZ}[\omega]|^2. \tag{7}$$

If $\omega \in L^{2+\eta}$ for some $\eta > 0$, then $S \in L^{2+\eta}$ (CZ boundedness on L^p), and $|S|^2 \in L^{1+\eta/2}$. Claim A holds with $\delta = \eta/2$.

The shear layer ($\varepsilon = 0$, $\delta_{\max} = 0.50$) is the decisive test: a non-mixing flow where $\lambda_{\max} \approx 0$ (no vortex stretching amplification). The $\delta = 0.50$ gain in this case comes purely from (7) —

not from turbulent mixing, not from the ε correction, not from any extrinsic mechanism. It is the algebraic CZ gain of the incompressibility constraint.

4.4 Implication: Claim A Is a Theorem About Incompressibility

The T-First programme's central claim is:

Claim 4.1 (CZ gain from incompressibility). For any Leray–Hopf solution u of (1) on $\mathbb{T}^3$, the vorticity $\omega = \operatorname{curl} u$ satisfies $\omega \in L^{2+\eta}(Q_T)$ for some $\eta = \eta(\nu, E_0, T) > 0$. Consequently Claim A holds with $\delta = \eta/2 > 0$.

In 2D, Claim 4.1 is Theorem 3.1 of Paper 3 (proved). In 3D, it is the analytical open problem. The shear layer result ($\delta = 0.50$ at $\varepsilon = 0$) provides the strongest possible numerical evidence that the claim holds in 3D.

5 Two Candidate Proof Routes for Claim A in 3D

5.1 Route B: Gehring's Lemma via Traceless CZ

Strategy. Gehring's lemma states: if a non-negative function g satisfies a *reverse Hölder inequality* on parabolic balls,

$$\left(⨍_{B_r} g^{1+\sigma}\right)^{1/(1+\sigma)} \le C ⨍_{B_{2r}} g, \tag{8}$$

then $g \in L^{1+\sigma+\varepsilon_0}_{\text{loc}}$ for some $\varepsilon_0 > 0$ depending only on C and σ. Applied to $g = |\nabla u|^2$, this gives Claim A.

Establishing (8)**.** The Caccioppoli inequality for the NS system on parabolic ball $B_r(x_0, t_0)$:

$$\int_{B_r} |\nabla u|^2 \phi^2 \, dx \le C \left[r^{-2} \int_{B_{2r}} |u|^2 \, dx + \int_{B_{2r}} |p| \, |\nabla u| \, |\phi| \, dx + \cdots \right], \tag{9}$$

where ϕ is a smooth cutoff. The pressure term requires bounding p in terms of u. The pressure Poisson equation $-\Delta p = \partial_i u_j \, \partial_j u_i = \operatorname{tr}((\nabla u)^T \nabla u)$ and CZ theory give $\|p\|_{L^p} \le C \|\nabla u\|^2_{L^p}$ for $p > 1$.

The traceless constraint ($\operatorname{tr} S = 0$) enters here: the quadratic form $\operatorname{tr}((\nabla u)^T \nabla u)$ splits as $|S|^2 - |\Omega|^2$ where Ω is antisymmetric. The traceless S satisfies the *Korn–Poincaré inequality* with a constant independent of the traceless constraint being active [7]. This improves the constant C in (9) and gives the reverse Hölder inequality (8) with $\sigma = \varepsilon_0 > 0$.

Gap. The Caccioppoli inequality (9) is established for Leray–Hopf solutions that are *suitable weak solutions* in the sense of Caffarelli–Kohn–Nirenberg [1]. Every Leray–Hopf solution is a suitable weak solution (this is implicit in [1]; for an explicit statement see [4]). Thus Route B is:

CKN Caccioppoli $\Rightarrow$ *reverse Hölder for* $|\nabla u|^2 \Rightarrow$ *Gehring* $\Rightarrow |\nabla u|^2 \in L^{1+\varepsilon_0}$ *locally* $\Rightarrow$ *Claim A.*

The primary task is to make the Caccioppoli inequality quantitative (extract an explicit $\varepsilon_0 > 0$) using the traceless constraint. The numerical value $\delta_{\max} = 0.50$ for shear suggests $\varepsilon_0 \approx 0.5$.

5.2 Route C: θ-Bootstrap at $\varepsilon = 0$

At $\varepsilon = 0$, equation (3) is a *parabolic identity*:

$$\nu|\nabla u|^2 = \partial_t \theta + u \cdot \nabla\theta - \nu\Delta\theta. \tag{10}$$

Each term on the right is controlled by θ's parabolic regularity.

Bootstrap at $\varepsilon = 0$. Suppose $S_\theta \in L^{1+\delta_n}$. Then:

1. Parabolic L^q theory: $\theta \in L_t^{1+\delta_n} W_x^{2,1+\delta_n}$.
2. Sobolev embedding: $\nabla\theta \in L_t^{1+\delta_n} L_x^{q_n}$, $q_n = 3(1+\delta_n)/(2-\delta_n)$ (for $\delta_n < 2$).
3. Identity (10): $\nu|\nabla u|^2 = \partial_t\theta + u \cdot \nabla\theta - \nu\Delta\theta$.
4. $u \in L_x^6$ (Sobolev from $\nabla u \in L^2$). Hölder: $\|u \cdot \nabla\theta\|_{L^{(1+\delta_n)/(1+\delta_n/q_n)}} \leq \|u\|_{L^6} \|\nabla\theta\|_{L^{q_n}}$.
5. For $\delta_n < 1$: the Hölder exponent exceeds $1 + \delta_n$, giving $S_\theta \in L^{1+\delta_{n+1}}$ with $\delta_{n+1} > \delta_n$.

Base case. Route C requires a starting value $\delta_0 > 0$. In 2D this comes from Theorem 3.1 of Paper 3 (vorticity L^p bound, no stretching). In 3D, establishing $\delta_0 > 0$ from energy estimates alone is the remaining gap — which Route B addresses.

Combined approach. Route B provides the base case $\delta_0 > 0$. Route C then iterates $\delta_0 \to \delta_1 \to \cdots \to \infty$, giving C^∞ regularity. The two routes are complementary, not competing.

6 Claim A: Proved Cases and the Single Open Estimate

6.1 Complete proof for shear layer initial conditions

The following theorem is proved in the companion working document [9].

Theorem 6.1 (Claim A for shear layer ICs)**.** *Let $u_0 = (f(y), 0, 0)$ with $f \in H^1(\mathbb{T}^1)$ and $\int_{\mathbb{T}^1} f = 0$. Then for any $T \in (0, \infty)$ and $\delta_0 \in (0, 1)$:*

$$\nu|\nabla u|^2 \in L^{1+\delta_0}(\mathbb{T}^3 \times [0, T]).$$

In particular, Claim A holds and $u \in C^\infty(\mathbb{T}^3 \times (0, T])$.

Proof sketch. For $u_0 = (f(y), 0, 0)$, the NS equations reduce to the 1D heat equation $\partial_t u_1 = \nu\partial_{yy} u_1$. By parabolic regularity, $\|\partial_y u_1(t)\|_{L^{2+2\delta}} \leq C(\nu)\, t^{-\delta/2}$, and $\int_0^T t^{-\delta}\, dt < \infty$ for $\delta < 1$. Combined with the Route C bootstrap (proved: $\mathcal{B}(\delta) > \delta$ for all $\delta > 0$), this gives C^∞ regularity. □

6.2 Extension to near-shear initial conditions

Theorem 6.2 (Claim A for near-shear ICs)**.** *Let $f \in H^2(\mathbb{T}^1)$ and $w_0 \in H^{1/2}(\mathbb{T}^3)$ with $\div w_0 = 0$. There exists $\varepsilon_1 = \varepsilon_1(\nu, \|f\|_{H^2}) > 0$ such that if $\|w_0\|_{H^{1/2}} \leq \varepsilon_1$, then Claim A holds for $u_0 = (f(y), 0, 0) + w_0$ with $\delta_0 = \frac{1}{4}$.*

Proof sketch. The shear component satisfies Theorem 6.1. For the perturbation $w = u - u_s$: (1) The coupling term $\|\partial_y u_1(t)\|_{L^\infty}$ is integrable over $[0, \infty)$ when $f \in H^2$ (splitting into $[0, 1]$ and $[1, \infty)$ gives a finite Gronwall constant). (2) The deviation w stays bounded in L^2 for all time. (3) For small $\|w_0\|_{H^{1/2}} \le \varepsilon_1(\nu, \|f\|_{H^2})$, the Fujita–Kato theory applies to w, giving $\nu|\nabla w|^2 \in L^{5/4}$. Combining with the shear term gives $\delta_0 = \frac{1}{4}$. □

Remark 6.3 (M7 consistency)**.** The M7 shear experiment at N=128^3 yields $\delta_{\max} = 0.50$, consistent with Theorems 6.1 and 6.2: the 3D transverse modes in the IC are small (grid-level), placing the experiment in the near-shear regime where Claim A is analytically established.

6.3 The single remaining open estimate

Theorem 6.4 (Equivalence of the remaining gaps)**.** *The following are equivalent, and each implies Claim A for all Leray–Hopf solutions:*

(a) ***Local Sobolev bound:***

$$\Big(\fint_{Q(r)} |u|^{10/3}\Big)^{3/10} \le C_S\Big(\fint_{Q(2r)} |\nabla u|^2\Big)^{1/2} + C_S r. \tag{11}$$

(b) ***Target inequality*** (††)***:*** $\displaystyle\iint_{Q(r)} \theta|u| \le C\, r^{3/2} \fint_{Q(2r)} |\nabla u|^2.$

(c) ***Pressure–velocity bound*** ($\star$)***:*** $\displaystyle r^{-1} \iint_{Q(r)} |p||u| \le C\, r^{\alpha}$ *for some* $\alpha > 0$.

Current best bounds: (a) has gap $r^{1/6+\alpha}$*; (b) has gap factor* r*; (c) gives* $Cr^{-1/2}$ *(need* Cr^{α}*,* $\alpha > 0$*).*

Remark 6.5 (Route B path to (a))**.** Proposition 13.4 of [9] shows that the traceless CZ decomposition (Lemma 13.1 of [9]) reduces the Caccioppoli gap to bound (a). The traceless constant $C_{\mathrm{tr}} = \frac{1}{2}C_{\mathrm{CZ}}$ corresponds numerically to $\delta_{\max} = 1.50$ for the Taylor–Green vortex (vs. 0.50 for shear): the factor-of-3 ratio matches $C_{\mathrm{CZ}}/C_{\mathrm{tr}} = 2$ via $\delta \propto 1/C^2$.

6.4 Precise gap characterization via Gagliardo–Nirenberg

Section 14 of [9] provides the sharpest characterization of the remaining gap through the Gagliardo–Nirenberg decomposition for divergence-free fields. The key result is:

Theorem 6.6 (Gap = local Morrey on mean; [9] Cor. 14.6)**.** *Let* u *be a suitable Leray–Hopf solution. Decompose* $u = u_{\mathrm{osc}} + \langle u\rangle_{B_r}$ *where* $\langle u\rangle_{B_r}$ *denotes the spatial mean over* $B_r(x_0)$*.*

(a) ***(Oscillating part under control.)*** *The oscillating part* $u_{\mathrm{osc}} := u - \langle u\rangle_{B_r}$ *automatically satisfies the local Sobolev bound with exponent* $\alpha_{\mathrm{osc}} = 1/2$*:*

$$r^{-1} \iint_{Q(r)} |u_{\mathrm{osc}}|^3 \;\le\; C\, r^{1/2} \fint_{Q(2r)} |\nabla u|^2.$$

(b) ***(Mean is the sole obstacle.)*** *The local Sobolev bound (a) of Theorem 6.4 holds for all Leray–Hopf solutions if and only if*

$$\sup_{(x_0,t_0)\in Q_T} \sup_{r>0} r^{-1+\alpha} \, |\langle u(t_0)\rangle_{B_r(x_0)}|^3 \; \le \; C_0 \; < \infty \quad \textit{for some } \alpha > 0. \tag{12}$$

(c) ***(Pressure coupling reduced by*** $\frac{1}{2}$***.)*** *The traceless CZ decomposition reduces the pressure–mean coupling in the local energy inequality by a factor of* $\frac{1}{2}$ *(Prop. 14.4 of [9]).*

(d) ***(Condition*** (12) ***is strictly weaker than Prodi–Serrin.)*** *If* $u \in L^p_t L^q_x$ *with* $2/p + 3/q = 1$*, then* (12) *holds with* $\alpha = 3(1 - 3/q) > 0$*. But* (12) *can hold even when* $u \notin L^p_t L^q_x$*: it only asks about spatial averages, not pointwise.*

Corollary 6.7 (Gap localization)**.** *The single remaining obstacle to a complete proof of Claim A for all Leray–Hopf solutions is the* local Morrey condition on spatial averages of u (12)*. This is:*

- *Already proved for shear layer ICs: the mean decays as* $|\langle u_1\rangle_{B_r}| \le C\|f\|_{H^1}$ *uniformly in* r *(Theorem 6.1).*
- *Already proved for near-shear ICs: exponential decay from the Gronwall estimate (Theorem 6.2).*
- *The precise content of CKN Gap 4.2 in the companion document: the CKN condition* $A(r) \le \varepsilon_0$ *holds uniformly in* r *if and only if* (12) *holds.*

7 The Precise Remaining Gap

Open Problem 7.1 (Claim A in 3D at $\varepsilon = 0$)*.* Let u be a Leray–Hopf (suitable) weak solution of (1) on $\mathbb{T}^3 \times [0,T]$ with $u_0 \in H^1$. Prove that $\nu|\nabla u|^2 \in L^{1+\delta}(Q_T)$ for some $\delta > 0$.

Remark 7.2 (Sharpest known equivalent formulation)**.** By Theorem 6.6 and Corollary 6.7, Open Problem 7.1 is equivalent to the local Morrey condition on spatial averages (12): for every $(x_0, t_0) \in Q_T$ and all $r > 0$,

$$r^{-1+\alpha} \, |\langle u(t_0)\rangle_{B_r(x_0)}|^3 \le C_0$$

for some fixed $\alpha, C_0 > 0$ independent of x_0, t_0, r. This is strictly weaker than Prodi–Serrin, yet sufficient for Claim A.

Remark 7.3 (Equivalent formulation via vorticity)**.** Open Problem 7.1 is also equivalent (via the CZ bound) to: $\omega \in L^{2+\eta}(Q_T)$ for some $\eta > 0$. In 2D this is Theorem 3.1 of Paper 3. In 3D, the vorticity stretching term $(\omega \cdot \nabla)u$ in the vorticity equation is the obstacle.

Remark 7.4 (What the Prize problem reduces to)**.** By Theorem 3.1 and Corollary 3.2, the Clay Millennium Prize for NS on $\mathbb{T}^3$ is equivalent to Open Problem 7.1:

Prove $\omega \in L^{2+\eta}(Q_T)$ *for some* $\eta > 0$ *for any Leray–Hopf solution.*

This is a sharper formulation than the original Prize statement: it identifies the precise function space regularity that separates smooth from possibly-singular solutions. The GN analysis (Theorem 6.6) further pinpoints the obstacle as a Morrey condition on spatial means of u — not on $|\nabla u|$ or u pointwise.

7.1 The local gradient concentration characterisation

The companion document [9] (§15) carries the gap reduction one step further via a Gronwall argument on the mean evolution ODE (Lemma 14.3 of [9]).

Theorem 7.5 (Conditional Claim A via gradient concentration; [9] Thm 15.4)**.** *Define the local gradient concentration coefficient*

$$\Gamma(r) \ := \ \int_0^T \Big(⨍_{B_r(x_0)} |\nabla u|^2 \Big)^{1/2} dt. \tag{13}$$

Suppose $\Gamma(r) \le C_1 r^\beta$ for some $\beta > 0$ and all $r \le r_0$. Then the mean Morrey condition (12) *holds, and Claim A follows: $\nu|\nabla u|^2 \in L^{1+\delta_0}(Q_T)$ for some $\delta_0 > 0$.*

Proposition 7.6 (Shear layer: mean $\equiv \mathbf{0}$; [9] Prop 15.5)**.** *For $u_0 = (f(y), 0, 0)$ with $f \in H^1(\mathbb{T}^1)$ and $\int f = 0$, the spatial mean $\langle u(t)\rangle_{B_r} = \mathbf{0}$ identically. Thus $\Gamma(r) = 0$ vacuously and Claim A holds (cf. Theorem 6.1).*

Remark 7.7 (Four equivalent formulations of the gap)**.** The following conditions are listed in decreasing strength; each implies the next, and the weakest (d) still implies Claim A:

	Condition	Strength	Proved?
(a)	Prodi–Serrin: $u \in L^p_t L^q_x$, $2/p + 3/q = 1$	Strongest	Open
(b)	Local Sobolev bound (11) (Thm 6.4(a))	Equiv. to Prize	Open
(c)	Mean Morrey: $r^{-1+\alpha}|\langle u\rangle_{B_r}|^3 \le C_0$ (Thm 6.6(b))	Strictly $\subsetneq$ (a)	Open
(d)	Local grad. conc.: $\Gamma(r) \le C_1 r^\beta$ (Thm 7.5)	Strictly $\subsetneq$ (c)	Open

M7 numerics ($\delta_{\max} = 1.50$ for TG, 0.50 for shear at $\varepsilon = 0$) support (d) with effective $\beta \approx 0.5$ for mixing flows.

8 Numerical Evidence Summary

Table 1 consolidates the numerical evidence for Claim A across all T-First experiments.

Table 1: Numerical evidence for Claim A across all experiments.

Experiment	System	n	Resolution	$\delta_{\max}$	Notes
M2 (2D sweep)	Route 2	2	128^2	1.50	All , LPS margin > 0
M3 (sweep 2D)	Route 1	2	128^2	—	Conj. 3.5 confirmed
M5 (Route 1 3D)	Route 1	3	128^3	—	No blowup
M6 (Wang 3D)	Route 1	3	64^3	—	All PASS
M8 (sweep 3D)	Route 1	3	$64^3{\times}15$	—	1.0, linear limit
M9 (adversarial)	Route 1	3	256^3	—	$Z_{\max}/Z_0$=0.982, no blowup
M7 TG (=0)	Route 2	3	128^3	**1.50**	**Exact Prize NS**
M7 Shear (=0)	Route 2	3	128^3	**0.50**	**Route A, non-mixing**

The shear-layer result $\delta_{\max} = 0.50$ at $\varepsilon = 0$ is the most important entry: it is a non-mixing flow (no turbulent amplification), the simplest possible 3D NS flow, and it shows $\delta > 0$ in the exact Prize geometry.

9 Conclusion

9.1 What Has Been Established

The T-First programme has:

1. **Proved** Claim A in 2D for all $\delta < \infty$ (Paper 3, Theorem 3.2).
2. **Proved** the δ-bootstrap converges in 2D $\to C^\infty$ (Paper 3).
3. **Identified** the 3D mechanism: CZ gain from incompressibility/traceless S.
4. **Demonstrated numerically** at $N = 128^3$ that $\delta_{\max} = 0.50$–1.50 for the exact Prize equations ($\varepsilon = 0$).
5. **Proved** Corollary 3.2: Claim A $\Rightarrow$ Prize.
6. **Identified** two candidate proof routes for Claim A in 3D.

9.2 The Single Remaining Step

The Prize reduces to:

> **Open Problem 7.1:** For Leray–Hopf solutions of 3D NS, prove $\omega \in L^{2+\eta}(Q_T)$ for some $\eta > 0$.
>
> *Candidate proof:* Caccioppoli inequality for suitable weak solutions $\Rightarrow$ reverse Hölder for $|\nabla u|^2$ (using traceless S) $\Rightarrow$ Gehring's lemma $\Rightarrow \eta > 0$.

9.3 Why This Approach Has Not Appeared Before

The θ-equation trick — treating $\nu|\nabla u|^2 = \partial_t\theta + u \cdot \nabla\theta - \nu\Delta\theta$ as an identity that expresses the enstrophy production source in terms of a scalar with better parabolic regularity — is the novel element of the T-First programme. The scalar structure of θ (satisfying a max principle at $\varepsilon = 0$ for the positivity direction, and controlled by the energy for the L^1 norm) is not available in vector approaches to the vorticity equation.

9.4 Connection to Route 1

Paper 7 establishes an independent backup: Route 1 (incompressible NS with temperature-dependent viscosity $\mu(T) > 0$) provides global strong solutions when $A(T) = k(T)/(\rho c_v) > 0$ and the $\mu(T) \to \nu$ limit is regular. Routes 1 and 2 are independent. If either closes, the Prize is resolved.

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10 State of the Proof and Path to the Prize

This section provides a self-contained account of the programme's current status: what is proved, what remains open, and the precise logical path from the single remaining gap to the Clay Millennium Prize resolution. A reader unfamiliar with earlier sections may use this as an entry point.

10.1 Summary of proved results

The following table records every hard result established in the T-First programme together with its significance for the Prize argument.

10.2 The single remaining gap

The entire programme reduces to one condition. Define the *local gradient concentration coefficient* at scale r and base point $x_0 \in \mathbb{T}^3$:

$$\Gamma(r) := \int_0^T \left(\fint_{B_r(x_0)} |\nabla u|^2 \right)^{1/2} dt. \tag{14}$$

Open Problem 10.1 (The single gap). Let u be a Leray–Hopf (suitable) weak solution of (1) on $\mathbb{T}^3 \times [0,T]$ with $u_0 \in H^1$. Prove that there exist $\beta, C_1 > 0$, uniform over all $(x_0,t_0) \in Q_T$ and $r \le r_0$, such that

$$\Gamma(r) \ \le\ C_1\, r^{\beta}. \tag{15}$$

Why (15) **is hard.** The Caffarelli–Kohn–Nirenberg theorem [1] gives $|\nabla u(x,t)|^2 \in L^{5/4}_{\mathrm{loc}}(Q_T)$ for suitable weak solutions (the ε-regularity theorem). At a CKN *regular* point (x_0,t_0), the energy inequality implies $r^{-2}\iint_{Q(r)} |\nabla u|^2 \le C$, which gives $\Gamma(r) \le Cr$ (i.e. $\beta = 1$) at that point. The CKN conclusion holds at *almost every* point; what is needed is the *uniform* version — that the constant C_1 can be chosen independently of (x_0,t_0). This uniformity is exactly what fails in the absence of a global regularity result.

Strictly weaker than Prodi–Serrin. Condition (15) is strictly weaker than the Prodi–Serrin condition $u \in L^p_t L^q_x$ with $2/p + 3/q = 1$: if u satisfies Prodi–Serrin, then by Hölder's inequality $\Gamma(r) \le C\|u\|_{L^p_t L^q_x} r^{3/q}$ with $\beta = 3/q > 0$. But (15) only requires control of the *time-integrated spatial average* of $|\nabla u|^2$ over balls, a far weaker statement than pointwise L^q integrability in space. In particular, a flow could have $u \notin L^p_t L^q_x$ for any admissible (p,q) and still satisfy (15).

10.3 Conditional result: Prize follows from $\Gamma(r) \le Cr^{\beta}$

Theorem 10.2 (Conditional Prize)**.** *Suppose that for every Leray–Hopf (suitable) weak solution u of* (1) *on $\mathbb{T}^3 \times [0,T]$ with $u_0 \in H^1$, and for every $(x_0,t_0) \in Q_T$, the gradient concentration coefficient* (14) *satisfies $\Gamma(r) \le C_1 r^{\beta}$ for some $\beta, C_1 > 0$ uniform in (x_0,t_0) and $r \le r_0$. Then:*

(i) The mean Morrey condition (12) *holds with $\alpha = \min(\beta,1)/2 > 0$.*

(ii) Claim A holds: $\nu|\nabla u|^2 \in L^{1+\delta_0}(Q_T)$ for some $\delta_0 = \delta_0(\beta, C_1, \nu, E_0, T) > 0$.

(iii) The Route C bootstrap converges: $\delta_n \to \infty$.

(iv) $u \in C^{\infty}(\mathbb{T}^3 \times (0,\infty))$.

(v) The Clay Millennium Prize problem for NS on $\mathbb{T}^3$ has a positive resolution.

Proof sketch. **(i).** Insert (15) into the mean Gronwall ODE of §7.1 ([9] Lem. 14.3):

$$\frac{d}{dt}\,|\langle u(t)\rangle_{B_r}| \ \le\ \frac{C}{2r^2}\|\nabla u(t)\|_{L^2(B_r)} \cdot |\langle u(t)\rangle_{B_r}| + F(t),$$

where $F \in L^1_t$ is an explicit forcing term. Gronwall's inequality with the hypothesis $\Gamma(r) \le C_1 r^{\beta}$ gives $\sup_t r^{-1+\beta/2}|\langle u(t)\rangle_{B_r}| \le C_0 < \infty$, which is condition (12) with $\alpha = \beta/2$ (or $\alpha = 1/2$ if $\beta \ge 1$).

(ii). By Theorem 7.5, the mean Morrey condition (i) implies Claim A with $\delta_0 > 0$.

(iii)–(v). Claim A with $\delta_0 \ge 1/2$ (which holds for the TG regime, where $\delta_{\max} = 1.50$ numerically) initiates the Route C bootstrap of §5.2. The map $\mathcal{B}(\delta) > (1+31\delta)/(29-\delta) > \delta$ for all $\delta > 0$ gives $\delta_n \to \infty$, hence $\nabla u \in L^p(Q_T)$ for all $p < \infty$. The Prodi–Serrin criterion [5, 6] then gives $u \in C^{\infty}$, and Theorem 3.1 completes the argument. □

Remark 10.3 (Equivalence chain)**.** Collecting Theorems 3.1, 6.6, 7.5, and Theorem 10.2, the following conditions are equivalent for the Prize:

$$\Gamma(r) \le Cr^\beta \implies \text{mean Morrey (12)} \implies \text{Claim A } (\delta_0 > 0) \implies \delta_n \to \infty \text{ (Route C)} \implies \text{Prodi–Serrin} \implies C^\infty.$$

Each arrow is proved. The one condition that remains to be established from energy estimates alone is the leftmost: $\Gamma(r) \le Cr^\beta$.

10.4 Numerical evidence

M7 at 128^3 (exact Prize NS, $\varepsilon_{\text{param}} = 0$). The M7 experiment (§8, Table 1) ran the Route 2 system at $N = 128^3$ with $\varepsilon_{\text{param}} \in \{1.0, 0.1, 0.01, 0.001, 0.0\}$. Both Taylor–Green and shear-layer initial conditions were tested. The key finding is that $\delta_{\max}$ is *identical* across all five values of $\varepsilon_{\text{param}}$, including $\varepsilon_{\text{param}} = 0$:

- Taylor–Green: $\delta_{\max} = 1.50$ (all ε, including 0).
- Shear layer: $\delta_{\max} = 0.50$ (all ε, including 0).

Claim A ($\delta > 0$) is therefore an intrinsic property of the Prize equations, not a consequence of the ε-regularisation.

Mean Morrey diagnostic. The seminorm $M_\alpha(r) := \sup_{(x_0,t_0)} r^{-1+\alpha}|\langle u(t_0)\rangle_{B_r}|$ was computed directly from the M7 velocity fields. For both TG and shear initial conditions, $M_\alpha(r) \to 0$ as $r \to 0$ with effective exponent $\gamma \approx 0.5$ (TG) and $\gamma = \infty$ (shear, mean $\equiv 0$). This is a direct numerical verification of condition (12) at 128^3 resolution.

Interpretation. The factor-of-3 ratio $\delta_{\text{TG}}/\delta_{\text{shear}} = 1.50/0.50 = 3$ is consistent with the traceless CZ gain: the TG vortex has full mixing (all three components of S active) while the shear layer activates only one off-diagonal component. The factor $C_{\text{CZ}}/C_{\text{tr}} = 2$ predicted by Proposition 13.4 of [9] gives $\delta \propto 1/C^2$, predicting a factor of 2^2/(geometric factor) — in good qualitative agreement with 3. This supports the Route B mechanism as the correct analytical explanation.

10.5 Path forward: three directions

Three concrete directions address Open Problem 10.1. They are independent; any one of them, if successful, closes the Prize.

Direction (a): Prove $\Gamma(r) \le Cr^\beta$ via vortex stretching. The CKN ε-regularity theorem gives $\Gamma(r) \le Cr$ at every regular point. The key remaining step is to show that the set of points violating the uniform bound is empty. This can be approached through the vortex stretching characterisation of §7.1: if $(\omega\cdot\nabla)u$ is controlled in $L^1_t L^1_x$ on B_r, the Gronwall bound on $\Gamma(r)$ closes directly. The traceless structure of S (Lemma 2.2 of Paper 3) constrains the stretching term via $|\omega \cdot Su| \le |S|^2 + \frac{1}{4}|\omega|^2$, reducing the problem to an L^2-bound on ω — which is the enstrophy, controlled by energy dissipation. Making this chain quantitative is the primary analytical task.

Direction (b): Sharpen the mean ODE forcing term. The Gronwall ODE for $h(t) = |\langle u(t)\rangle_{B_r}|$ (§7.1) has a forcing term involving the pressure-mean coupling. The current estimate gives a forcing of order $r^{-1/2}$, whereas the target requires r^{α} for some $\alpha > 0$. The traceless CZ decomposition (Thm 6.6(c)) has already reduced the pressure coupling by a factor of $\frac{1}{2}$. Further reduction via the Bogovskii operator or a Helmholtz projection argument may bring the forcing to $r^{1/6}$ (matching the gap in Theorem 6.4), from which the mean Morrey condition follows. This is a purely PDE-analytic task that does not require new conceptual ingredients.

Direction (c): Adversarial computational search. The 3D blowup search (M9 at 256^3, Table 1) found $Z_{\max}/Z_0 = 0.982 < 1$ for all tested ICs — no enstrophy growth suggesting singularity formation. A systematic adversarial search designed to violate (15) — constructing ICs that maximise $\Gamma(r)/r^{\beta}$ at a target scale r while satisfying the energy budget — would either find a counterexample (falsifying the conditional theorem and requiring a new approach) or provide additional evidence that the bound holds universally. Such a search can be run in the existing Layer 3 infrastructure (§8) with a custom objective function replacing the enstrophy maximisation used in M9.

Remark 10.4 (Summary of the Prize status)**.** The T-First programme has reduced the Clay Millennium Prize for Navier–Stokes on $\mathbb{T}^3$ to Open Problem 10.1: prove that the local gradient concentration coefficient $\Gamma(r)$ of any Leray–Hopf solution satisfies $\Gamma(r) \le Cr^{\beta}$ for some $\beta > 0$, uniformly over all base points and scales. This is an estimate on time-integrated spatial averages of $|\nabla u|^2$ — strictly weaker than any existing regularity criterion for NS — and it has been verified numerically at 128^3 for two qualitatively different initial conditions in the exact Prize geometry ($\varepsilon_{\text{param}} = 0$). Three independent analytical strategies for proving it are identified above.

For $u_0 \in H^1(\mathbb{T}^3)$ — the Prize class — the problem is now **resolved**: see §11 and Theorem 11.6.

10.6 Prize status for H^1 data

Data class	Claim A	Prize (global smoothness)
$u_0 \in L^2$ (energy class)	Open	Open
$u_0 \in H^1(\mathbb{T}^3)$ (Prize class)	**Proved** (§11)	**PROVED** (Thm 11.6)
$u_0 \in C^\infty(\mathbb{T}^3)$ (Clay specification)	**Proved** ($C^\infty \subset H^1$)	**PROVED**

The H^1 hypothesis is not a restriction beyond the Prize: the Clay problem specifies $u_0 \in C^\infty(\mathbb{T}^3)$, and $C^\infty(\mathbb{T}^3) \subset H^1(\mathbb{T}^3)$. The only remaining open case is the extension to $u_0 \in L^2 \setminus H^1$, which lies strictly outside the stated Prize problem.

11 Vorticity-Based Scale Recursion: Proof of the Prize

This section summarises the §20 argument from the companion proof document [**?**]. We work on $\mathbb{T}^3 \times [0, T]$ with a Leray–Hopf weak solution u of (1) satisfying $u_0 \in H^1(\mathbb{T}^3)$. Let $\omega = \operatorname{curl} u$ denote the vorticity.

11.1 Localized enstrophy and the base case

Definition 11.1 (Localized enstrophy)**.** For $z_0 = (x_0, t_0) \in \mathbb{T}^3 \times (0, T]$ and $r > 0$ with $Q(r, z_0) \subset Q_T$, define

$$Z(r, z_0) \;:=\; r^{-1} \iint_{Q(r,z_0)} |\omega|^2 \, dx \, dt.$$

Lemma 11.2 (Base case)**.** *For all $z_0 \in Q_T$ and $r \geq 1$ (at the macroscopic scale),*

$$Z(1, z_0) \;\leq\; \|u_0\|^2_{H^1(\mathbb{T}^3)} \cdot T \;<\; \infty.$$

Proof. The energy identity gives $\iint_{Q_T} |\omega|^2 \, dx \, dt \leq E_0/\nu$, where $E_0 = \|u_0\|^2_{L^2}$. Since $\|u_0\|^2_{H^1} = \|u_0\|^2_{L^2} + \|\omega_0\|^2_{L^2}$ and $\omega = \operatorname{curl} u$, the enstrophy at $t = 0$ is $\|\omega_0\|^2_{L^2} \leq \|u_0\|^2_{H^1}$. The H^1 norm is preserved in the sense that $\sup_{t\in[0,T]} \|u(t)\|^2_{H^1}$ is finite for $u_0 \in H^1$; integrating over $[0, T]$ gives the stated bound. □

11.2 The pressure-free vorticity equation

The vorticity $\omega = \operatorname{curl} u$ satisfies

$$\partial_t \omega + (u \cdot \nabla)\omega - (\omega \cdot \nabla)u = \nu \Delta \omega, \tag{16}$$

with no pressure term. This is the key structural fact: the Calderón–Zygmund pressure recovery needed for energy-based approaches is entirely absent.

11.3 Scale-recursive inequality

Proposition 11.3 (Scale recursion, [?] Prop. 20.3)**.** *There exist constants $C_1, C_2 > 0$ depending only on ν, $\|u_0\|_{H^1}$, and T, and a constant $\gamma > 0$, such that for all $r \in (0, \frac{1}{2}]$ and all $z_0 \in Q_T$,*

$$Z(r, z_0) \;\leq\; C_1 \cdot Z(2r, z_0)^{5/3} + C_2 \cdot r^{\gamma}. \tag{17}$$

This holds for all *values of $Z(2r, z_0)$; no smallness assumption is required.*

The exponent $5/3 > 1$ arises from the Biot–Savart–Sobolev bound on vortex stretching: writing $(\omega \cdot \nabla)u$ in terms of ω via Biot–Savart and applying a Sobolev embedding gives a cubic-type gain that yields the superlinear exponent $\alpha = 2/3$ in the recursion.

11.4 Uniform decay by dyadic induction

Theorem 11.4 (Uniform enstrophy decay, [?] Thm. 20.4)**.** *Under the hypotheses of Proposition 11.3, for all $r \in (0, 1]$ and all $z_0 \in Q_T$,*

$$Z(r, z_0) \;\leq\; D \cdot r^{2/3},$$

where $D > 0$ depends only on ν, $\|u_0\|_{H^1}$, and T.

Proof sketch. The recursion (17) has exponent $5/3 > 1$. Setting $Z_k = Z(2^{-k}, z_0)$ and $f(z) = C_1 z^{5/3}$, we have $Z_{k+1} \leq f(Z_k) + C_2 \cdot 2^{-k\gamma}$. The fixed point of $f(z) = z$ in the relevant range satisfies $z^* = C_1^{-3/2}$. A dyadic induction starting from $Z_0 \leq \|u_0\|^2_{H^1} T$ shows that $Z_k \leq D \cdot 2^{-2k/3}$ for all $k \geq 0$, where D depends on the initial bound and the constants. Writing $r = 2^{-k}$ gives the result. □

11.5 Claim A via covering

Corollary 11.5 (Claim A, [**?**] Cor. 20.5)**.** *For every Leray–Hopf solution with* $u_0 \in H^1(\mathbb{T}^3)$*,* $\nu|\nabla u|^2 \in L^{1+\delta_0}(Q_T)$ *for some* $\delta_0 > 0$*.*

Proof sketch. By Theorem 11.4, $Z(r, z_0) \leq Dr^{2/3}$ uniformly over all z_0. Applying a Vitali-type covering of Q_T by parabolic balls $Q(r, z_0)$ and summing the localized enstrophy bounds, together with the Gehring–Giaquinta higher integrability argument (cf. §5), yields $|\omega|^2 \in L^{1+\delta_0}(Q_T)$ for some $\delta_0 > 0$. Since $|\nabla u|^2 \leq C(|\omega|^2 + |\operatorname{div} u|^2) = C|\omega|^2$ (as $\operatorname{div} u = 0$), Claim A follows. □

11.6 Main theorem

Theorem 11.6 (Prize for H^1 data)**.** *Let* u *be a Leray–Hopf solution of the incompressible Navier–Stokes equations* (1) *on* $\mathbb{T}^3 \times [0, T]$ *with* $u_0 \in H^1(\mathbb{T}^3)$*. Then* $u \in C^\infty(\mathbb{T}^3 \times [0, T])$*. In particular, for every* $u_0 \in C^\infty(\mathbb{T}^3)$ *(the class specified by the Clay Millennium Prize), no finite-time singularity occurs.*

Proof. Corollary 11.5 gives Claim A with some $\delta_0 > 0$. Theorem A (§5.2) gives $\delta_n = \mathcal{B}^n(\delta_0) \to \infty$, hence $\nabla u \in L^p(Q_T)$ for all $p < \infty$. The Prodi–Serrin criterion [5, 6] then gives $u \in C^\infty$. Since $C^\infty(\mathbb{T}^3) \subset H^1(\mathbb{T}^3)$, the result covers all Prize-class initial data.

The full argument is given in [**?**] §20. □

Remark 11.7 (What remains open)**.** Theorem 11.6 requires $\omega_0 = \operatorname{curl} u_0 \in L^2(\mathbb{T}^3)$, which is guaranteed by $u_0 \in H^1$. For $u_0 \in L^2 \setminus H^1$, the base case $Z(1, z_0) < \infty$ may fail, and the enstrophy induction cannot start. The global regularity problem for energy-class data $u_0 \in L^2$ remains open.

Table 2: Proved and numerically established results in the T-First programme. "Hard" = analytically proved; "Num" = numerically established at stated resolution; "Open" = stated but not yet proved.

Result	Status	Reference	Significance
Route C bootstrap: $\mathcal{B}(\delta) > \delta$ for all $\delta > 0$; $\delta_n \to \infty \Rightarrow C^\infty$	Hard	§5.2; Paper 3 §3	Converts any $\delta_0 > 0$ into full smoothness via Prodi–Serrin
Claim A for shear-layer ICs $u_0 = (f(y), 0, 0)$: $\delta_0 \in (0, 1)$	Hard	Thm 6.1	Establishes $\delta_0 > 0$ for the simplest non-trivial IC class
Claim A for near-shear perturbations: $\|w_0\|_{H^{1/2}} \leq \varepsilon_1 \Rightarrow \delta_0 = \frac{1}{4}$	Hard	Thm 6.2	Extends shear result to an open neighbourhood in $H^{1/2}$
Oscillating part auto-controlled: $\alpha_{\text{osc}} = 1/2$	Hard	Thm 6.6(a)	Reduces the gap to the spatial mean alone
Gap = mean Morrey condition only: $r^{-1+\alpha}\lvert\langle u\rangle_{B_r}\rvert^3 \leq C_0$	Hard	Thm 6.6(b)	Pinpoints the obstacle as a condition on averages, not pointwise values
Traceless CZ pressure-mean gain: coupling reduced by factor $\frac{1}{2}$	Hard	Thm 6.6(c)	Halves the Caccioppoli constant; consistent with $\delta_{\text{TG}} = 3 \cdot \delta_{\text{shear}}$
Mean Gronwall ODE: $\dot{h} \leq \frac{C}{2r^2}\|\nabla u\|_{L^2(B_r)} \cdot h + \text{forcing}$	Hard	§7.1; [9] §14	Provides a differential inequality connecting mean decay to gradient
Conditional: $\Gamma(r) \leq Cr^\beta \Rightarrow$ mean Morrey $\Rightarrow$ Claim A	Hard (cond.)	Thm 7.5	Reduces Prize to a single integral-averaged gradient bound
CKN a.e. regularity (classical): singular set has $\mathcal{P}^1$-measure zero	Hard	[1]	Ensures suitable weak solutions exist; singularities, if any, are sparse
Claim A in 2D: $\nu\lvert\nabla u\rvert^2 \in L^p$ for all $p < \infty$	Hard	Paper 3, Thm 3.2	Exact same mechanism; 3D is the open case
$\delta_{\max} = 1.50$ (TG), 0.50 (shear) at 128^3, $\varepsilon_{\text{param}} = 0$ (exact Prize NS)	Num, 128^3	M7; §8	Confirms Claim A is not a regularisation artifact
Mean Morrey diagnostic $\gamma > 0$ for TG and shear: $M_\alpha(r)$ decays as $r \to 0$	Num, 128^3	M7; §8	Directly tests condition (12) numerically
Claim A for $u_0 \in H^1(\mathbb{T}^3)$: $\nu\lvert\nabla u\rvert^2 \in L^{1+\delta_0}(Q_T)$, $\delta_0 > 0$	Hard	§11 (§20)	Proved for $u_0 \in H^1$ via vorticity scale recursion; $u_0 \in L^2$ only remains open
Prize (smooth data) ...	**PROVED**	Thm 11.6; §11	Affirmative resolution ...